Prior Art Search Report

ARICCA-X: ACADEMIC RESEARCH INTEGRITY & CREDIBILITY CORRELATION ANALYZER: DETERMINISTIC, MACHINE-EXECUTABLE MULTI-SIGNAL SYSTEM FOR ACADEMIC INTEGRITY VERIFICATION

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SECTION 1A - Introduction

The present invention, ARICCA-X (Academic Research Integrity & Credibility Correlation Analyzer), is a deterministic, machine-executable data processing system for multi-signal verification of academic integrity. It addresses predatory and hijacked publication venues, citation cartels, AI-generated peer reviews and failures to comply with funding mandates. Existing tools address these problems separately, and many rely on subjective human curation or on machine learning models whose decisions cannot be fully audited. ARICCA-X instead evaluates all four problems in one pipeline. Every result is produced by an explicit formula, graph algorithm or Boolean rule, so each output carries a human-readable mathematical justification.

The system receives four input streams: venue data, a citation network, peer-review text with the reviewer's recommendation, and publication metadata with the applicable funding mandates. Each stream is processed by a dedicated module.

The first module, Venue Intelligence, evaluates the credibility of a conference or journal. It computes the interval between the announcement of a call for papers and its submission deadline, and raises a "Predatory Urgency" flag when this interval is shorter than 30 days. It also computes the Levenshtein edit distance between the submitted conference URL and known legitimate registry URLs, such as IEEE and ACM, to detect spoofed or hijacked venues. In addition, it applies rule-based tense analysis to the call-for-papers text to detect plagiarised or improperly reworded calls.

The second module, Researcher Auditing, analyses a directed citation graph in which nodes represent authors and edges represent citations. It runs a bounded depth-first search, with a maximum depth of 5 in the preferred embodiment, to find reciprocal citation loops among authors that standard citation indices do not detect. The module also divides the researcher's citations into self-citations, co-author citations and independent citations. It then recalculates the h-index using independent citations only, giving a True Independent H-Index. Finally, it computes a Citation Independence Ratio, defined as independent citations divided by total citations, and raises an inflation alert when this ratio falls below 0.30.

The third module, Peer-Review Forensics, analyses peer-review text using deterministic linguistic fingerprinting instead of a probabilistic classifier. It computes the coefficient of variation of sentence lengths and flags a review as artificially uniform when this value falls below 0.20. It measures hedging density by matching the text against a predefined list of hedging phrases. It computes a specificity score by using regular expressions to count explicit references to manuscript elements. It also computes a sentiment polarity score, and raises a misalignment flag when the review text is strongly positive but the reviewer's recommendation is Reject.

The fourth module, Grant Compliance, holds machine-readable Boolean rule matrices for funding mandates such as NIH, NSF, Plan S and UKRI. It evaluates a publication's metadata, including embargo length, open access type, licence and repository status, against every rule

of the applicable mandate. For each rule it produces a pass or fail record with a remediation note, giving an audit trail for each publication.

A cross-correlation engine receives the discrete flags, metrics and pass/fail records produced by the four modules and aggregates them into a composite risk profile for the venue, researcher or publication under review. Every intermediate value is produced by an explicit formula or Boolean rule. The composite risk profile is therefore delivered with a human-readable mathematical justification and a verifiable evidence chain, and no probabilistic classifier is used in the core pipeline.

The invention further discloses a supplementary credibility logic engine with four mechanisms. Adaptive Decay Weighting (ADW) handles missing evidence. When some credibility dimensions are absent, their predetermined weights are redistributed proportionally over the dimensions that are present ($w'i = wi / \Sigma k \in A\ wk$), and the score is computed as $\Sigma\ w'i \times si$. An absent dimension is treated as unknown rather than given a fabricated score. Cross-Signal Anomaly Detection (CSAD) normalises the independent signal vectors and computes the mathematical divergence D between them. When D exceeds a threshold τ, CSAD amplifies the risk proportionally ($Risk' = Risk \times (1 + \alpha \cdot g(D, \tau))$), without using a machine learning classifier. Grammatical Tense Analysis (GTA) uses regular expressions to classify a venue's indexing statements into present, past, future and conditional categories, and future and conditional statements carry a higher deterministic penalty than present and past statements. Temporal Fingerprint Evolution Tracking (TFET) compares multi-dimensional fingerprints of a venue taken at two times, T1 and T2. It computes the drift vector $\Delta F = F(T2) - F(T1)$ and raises an alert when the magnitude of ΔF exceeds an operational stability threshold.

These mechanisms run in a six-phase deterministic pipeline: component scoring, Boolean heuristics, credibility computation using ADW and CSAD, risk classification, rule-based recommendation, and flag generation. The state after each phase is strictly serializable, so the evidence and decision state can be audited at every phase.

The system uses a bifurcated architecture. Machine learning models are used only in an isolated offline environment for research and validation. Human operators translate their findings into deterministic rules, and the production pipeline delivered to users contains no machine learning model. The system is implemented as a set of software modules, a web application and a command-line interface. Its developers report that it processed 11,500 test inputs in automated stress testing without error.

SECTION 1B - KEY FEATURES OF THE INVENTION

A deterministic, machine-executable data processing system for multi-signal academic integrity verification, comprising:

1. Automated venue credibility assessment (temporal integrity of the call for papers and edit-distance detection of spoofed or hijacked venue URLs)
2. Citation-network auditing for citation-cartel/ring detection and h-index decomposition with citation-independence analysis
3. Rule-based linguistic fingerprinting of peer-review text (sentence-length variation, hedging density and sentiment against reviewer recommendation)
4. Rule-based verification of publication metadata against funding mandates with a pass/fail result per rule
5. Multi-signal correlation of module outputs into a composite credibility/risk profile with an auditable evidence record

The invention also includes a supplementary credibility logic engine comprising Adaptive Decay Weighting (ADW), Cross-Signal Anomaly Detection (CSAD), Grammatical Tense Analysis (GTA), Temporal Fingerprint Evolution Tracking (TFET) and a six-phase deterministic processing pipeline. These mechanisms have also been considered in the prior art search where relevant.

Relevant References:

Patents / Literature	**1**	**2**	**3**	**4**	**5**
1. US9178901B2	√				
2. US12113828B2	√				√
3. US7493403B2	√				√
4. AU2012231158B2			√		√
5. US12361214B2			√		
6. Schreiber (2007)		√			
7. Kojaku et al. (2021)		√			
8. Spanakis et al. (2026)	√	√			√
9. Baidya et al. (2026)			√		
10. cOAlition S Journal Checker Tool (2020)				√	√

Keywords:

- Research integrity / academic integrity / publication integrity verification / research misconduct detection
- Citation network analysis / citation graph / directed citation graph / author citation network
- Citation cartel / citation ring / citation stacking / anomalous citation group
- Reciprocal citation / coordinated citation / citation clique / graph cycle detection / depth-first search
- H-index decomposition / self-citation correction / co-author citation / sharpened h-index / citation independence
- AI-generated text detection / machine-generated text / peer-review authenticity / rule-based text analysis
- Linguistic fingerprinting / stylometric features / sentence-length variance / hedging density / sentiment polarity
- Predatory journal / predatory conference / hijacked journal / venue credibility assessment
- Typosquatting / spoofed domain / URL edit distance / Levenshtein distance
- Funder mandate / open access compliance / Plan S compliance checker / Boolean rule engine
- Credibility score / composite risk score / multi-signal correlation / multi-source score aggregation
- Deterministic decision system / auditable evidence chain / explainable decision record / audit trail
- Missing data handling / missing signal / weight redistribution / available-components scoring
- Cross-signal anomaly detection / signal divergence / risk amplification
- Temporal fingerprint / registration drift / domain ownership change / WHOIS monitoring
- Grammatical tense analysis / indexing claim verification / tense classification

In addition to manual and keyword searching in the relevant subclasses, keyword searching was conducted with no class limitation.

SECTION 2A - Search Results

Results of the search - Relevant

1. US9178901B2 MALICIOUS UNIFORM RESOURCE LOCATOR DETECTION

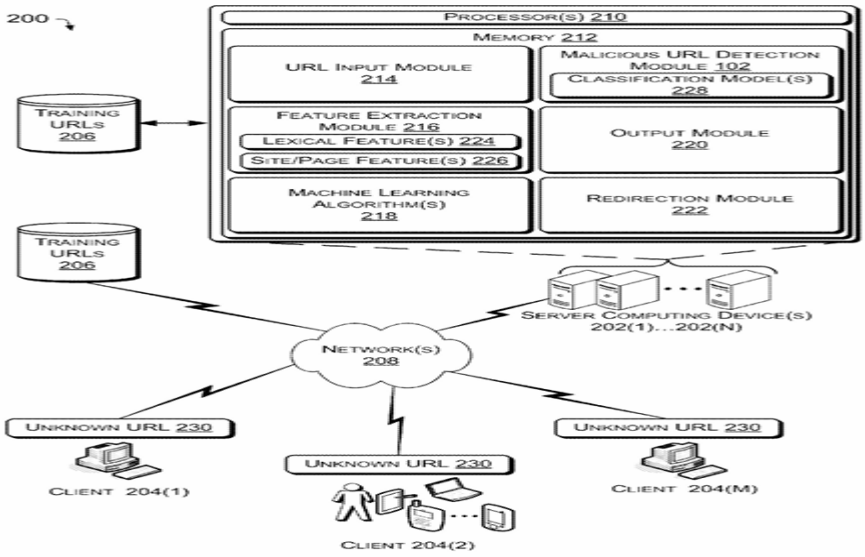
[ABSTRACT]

The techniques described herein use training data to train classification models to detect malicious Uniform Resource Locators (URLs) that target authentic resources (e.g., Web page, Web site, or other network locations accessed via a URL). The techniques train the classification models using one or more machine learning algorithms. The training data may include known benign URLs and known malicious URLs (e.g., training URLs) that are associated with a target authentic resource. The techniques then use the trained classification models to determine whether an unknown URL is a malicious URL. The malicious URL determination may be based on one or more lexical features (e.g., brand name edit distances

for a domain and path of the URL) and/or site/page features (e.g., a domain age and a domain confidence level) extracted.

[DETAILED DESCRIPTION]

The document describes a malicious URL detection module 102 in which a feature extraction module 216 extracts lexical features 224 and site/page features 226 from a URL. The lexical features include a brand name edit distance, computed as the minimum edit distance between the substrings of the URL and a set of known brand names B (Equations 1 and 2), separately for the domain and the path. The site/page features include a domain age obtained from a domain registry, a domain confidence level computed from the ratio of known benign to known malicious URLs hosted by the second level domain (Equation 3), and link popularity. Support vector machine classification models 228 trained on labelled URLs classify an unknown URL as malicious and identify the legitimate entity it targets. A redirection module 222 first resolves redirected URLs to their true URLs, and an output module 220 warns the user or blocks the URL (FIGS. 4 and 5).



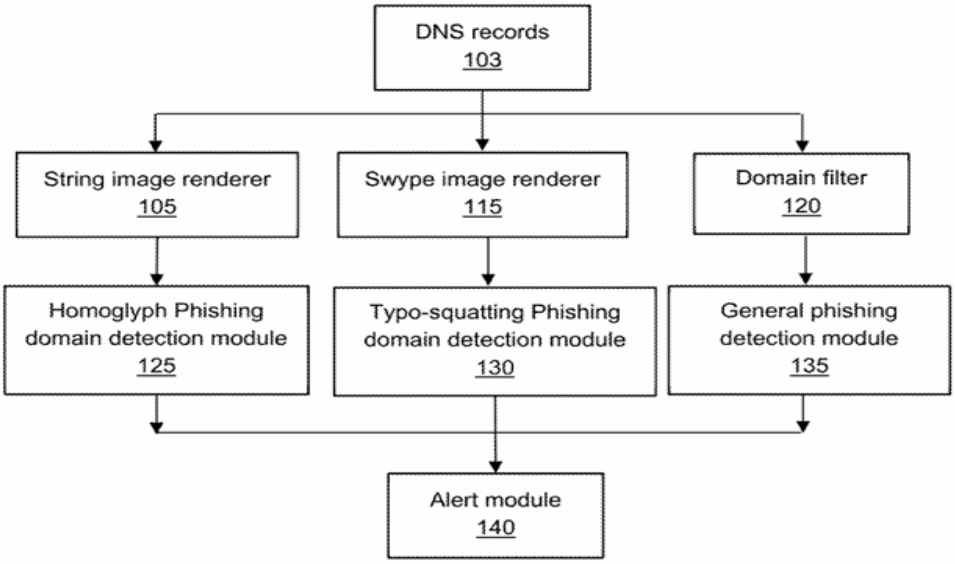
2. US12113828B2 SYSTEM AND METHOD FOR DETECTING PHISHING-DOMAINS IN A SET OF DOMAIN NAME SYSTEM (DNS) RECORDS

[ABSTRACT]

This document describes a system and method for detecting phishing-domains, which are used by cyber-attackers to carry out phishing attacks, in a set of Domain Name System (DNS) records, the system comprising a homoglyph phishing domain detection module, a typo-squatting phishing domain detection module, a general phishing domain detection module and an alert module. These modules are configured to collaboratively detect and identify phishing-domains from the set of DNS records using a combination of homoglyph, typo-squatting and general phishing domain techniques. Subsequently, an alert module may be used to correlate the alerts from the various phishing detection modules to discover phishing campaigns occurring in DNS network data.

[DETAILED DESCRIPTION]

The document describes three detection modules feeding a common alert module. The homoglyph module renders punycode domains as string images, encodes them with a trained encoder and flags a domain when the Euclidean distance to any encoding in a reference database falls below a predetermined threshold. The typo-squatting module renders each domain as a Swype-like keyboard path image and classifies it in the same way, and the description also implements a weighted Damerau-Levenshtein edit distance, weighted by keyboard distance, between the queried domain and the targeted legitimate domain. The general module scores domains that partially match a phishing reference list with a transformer network and resolves their IP addresses and associated external domains. The alert module correlates the alerts of all modules to identify phishing campaigns.



3. US7493403B2 DOMAIN NAME OWNERSHIP VALIDATION

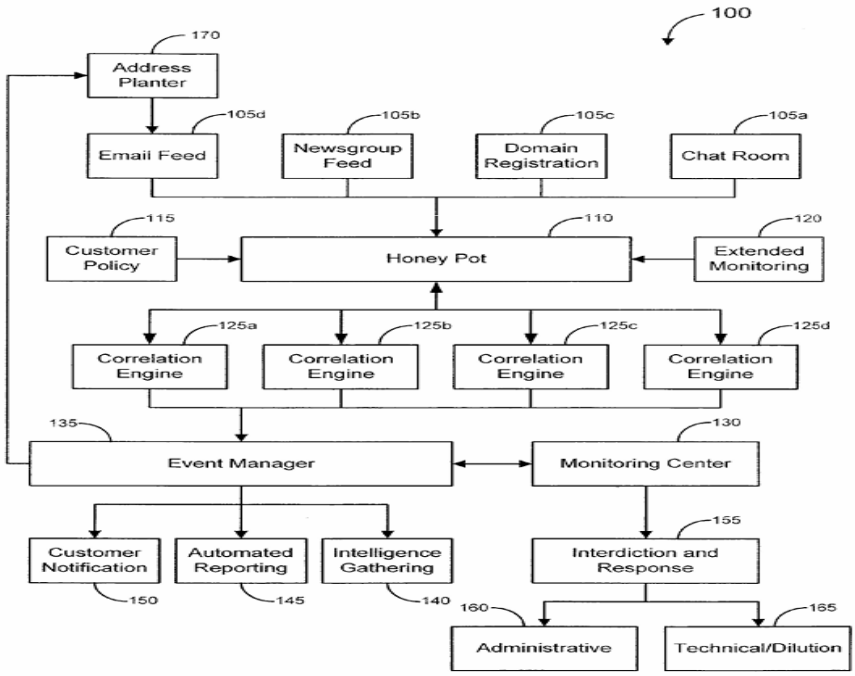
[ABSTRACT]

Embodiments of the invention provide systems and methods for validating ownership of a domain name. According to one embodiment, a validating ownership of a domain name can comprise retrieving one or more domain name ownership records. For example, the one or more domain name ownership records comprise Who Is records. Validity of the one or more domain name records can be confirmed with a designated domain manager. According to one embodiment, confirming validity of the one or more domain name records with the designated domain manager can comprise authenticating the designated domain manager based on a certificate provided by the designated domain manager.

[DETAILED DESCRIPTION]

The document describes a fraud detection system 100 (FIG. 1A) in which data sources, including chat rooms, newsgroups, domain registration zone files and e-mail feeds, feed a honey pot 110 that passes data to a plurality of correlation engines 125a to 125d. The

correlation engines signal an event manager 135, which drives investigation, automated reporting, customer notification and interdiction. Zone files are scanned for registrations similar to a customer's domain, such as other top level domains and similar spellings, and suspicious domains are placed on a monitoring list. An authentication registry 405 runs a validation process 408 (FIG. 6) that harvests WHOIS information 615 and domain information 625, confirms them with a designated domain manager and stores a validity flag or score in a registration database 407 and a reputation service 420. The ownership records are then monitored, and a domain is re-validated, or its validated status revoked, when its ownership records change (FIG. 7).



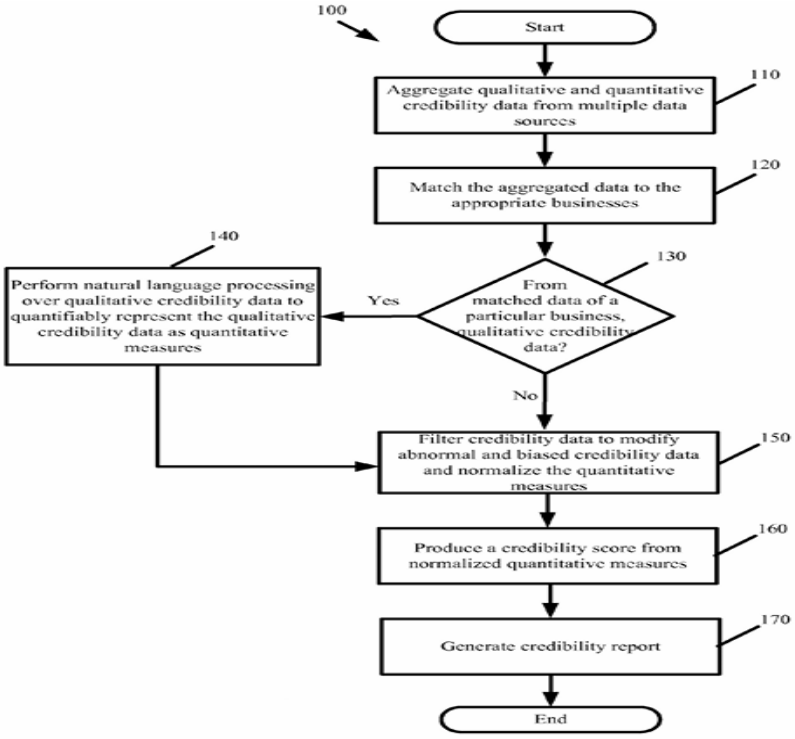
4. AU2012231158B2 CREDIBILITY SCORING AND REPORTING

[ABSTRACT]

The present document discloses a method for producing a score to quantifiably represent credibility of a particular entity, the method comprising aggregating from a plurality of data sources, credibility data comprising (i) qualitative data having a textual statement pertaining to each entity of a first plurality of entities and (ii) quantitative data having a quantitative measure for quantifiably rating each entity of a second plurality of entities, processing the qualitative data to derive a set of quantitative measures for quantifiably rating each entity of the first plurality of entities based on text of a textual statement pertaining to that entity, normalizing the set of quantitative measures derived from the qualitative data and the quantitative measures from the quantitative data, and producing a credibility score for a particular entity of the first and second plurality of entities based on normalized quantitative measures that relate to the particular entity.

[DETAILED DESCRIPTION]

The document describes a credibility scoring system that aggregates credibility data from multiple data sources through plug-in modules and matches it to a particular business. Natural language processing identifies, in each textual statement, a first word connoting a degree of positivity or negativity and a second word modified by it, and maps the pair to a scale of values. Scoring filters normalise the measures with disproportionate weighting, giving more weight to more trusted data sources and less weight to stale data. A credibility scoring aggregator 640 combines the normalised measures by averaging, summing or proprietary formulas, so that the score can be computed with any number of available measures. A reporting engine and interface portal 240 present the score with drill-down layers.



5. US12361214B2 COMPUTER-IMPLEMENTED METHOD FOR DISTRIBUTIONAL DETECTION OF MACHINE-GENERATED TEXT

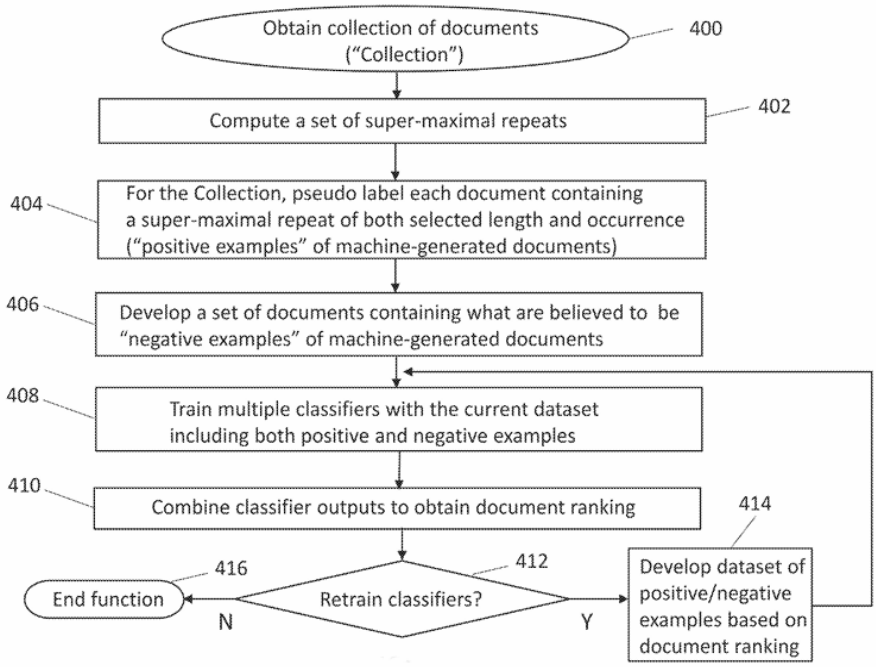
[ABSTRACT]

There is disclosed a computer-implemented method for detecting machine-generated documents in a collection of documents including machine-generated and human-authored documents. The computer-implemented method includes computing a set of long-repeated substrings (such as super-maximal repeats) with respect to the collection of documents and using a subset of the long-repeated substrings to designate documents containing the subset of the repeated substrings as machine-generated. The documents designated as machine-generated serve as positive examples of machine-generated documents and a set of documents including at least one human-authored document serves as negative examples of machine-generated documents. A plurality of classifiers are trained with a dataset including both the positive and negative examples of machine-generated documents. Classified output of the

classifiers is then used to detect an extent to which a given document of the dataset is machine-generated.

[DETAILED DESCRIPTION]

The document describes computing super-maximal repeats over a document collection with a suffix tree and pseudo-labelling each document containing a repeat of at least 20 characters occurring at least 3 times as machine-generated (FIG. 4). Negative examples are drawn at random from the collection or from a separate human-authored set, an ensemble of binary classifiers is trained, and their predictions are summed to rank documents by the extent to which they are machine-generated, with optional retraining on the ranking. In an alternative embodiment (FIG. 5), the natural language processing feature is selected from lexical, stylometric, graphemic and syntactic classes, such as word length, sentence length, paragraph length or counts of short words, and training sets are built from documents above and below thresholds of that feature.



6. A CASE STUDY OF THE HIRSCH INDEX FOR 26 NON-PROMINENT PHYSICISTS

[ABSTRACT] (summary)

The author computes the h-index for 26 physicists who are not highly prominent, so that the results reflect the average scientist rather than leading ones. The study examines quantitatively the pitfalls in determining the index from the Web of Science and the effect of self-citations on its value.

[DETAILED DESCRIPTION]

The paper defines three kinds of self-citation correction. The first kind removes the author's own self-citations, the second kind removes the highest self-citation count of any single co-author, and the third kind removes the cumulative self-citations of all co-authors, found by checking every citing paper for co-authorship. Papers are re-sorted by the corrected citation counts, and the h-index recomputed on those counts gives a sharpened index hs. Table 2 reports the corrected indices and the ratios between the corrected and original indices for all 26 data sets, an average reduction of 21.3%. The author notes that hs can still be inflated by reciprocal citing between colleagues.

data set	h	h_o	h_c	h_s	$\hat{r}$	$\underline{r}$	$\bar{r}$	$\tilde{h}_o/\tilde{h}$	$\tilde{h}_c/\tilde{h}$	$\tilde{h}_s/\tilde{h}$	order($\tilde{h}_s$)
A	39	36	34	34	46	40	22	0.92	0.88	0.87	1
B	27	24	23	22	34	28	15	0.88	0.85	0.80	2
C	23	22	21	20	29	29	13	0.93	0.91	0.87	3
D	20	17	16	16	34	34	7	0.85	0.83	0.80	4
E	19	16	15	15	22	17	12	0.83	0.79	0.79	5
F	18	17	14	14	21	18	11	0.94	0.78	0.78	6
G	17	11	11	11	21	15	10	0.65	0.65	0.65	13
H	16	14	13	13	18	15	13	0.91	0.85	0.83	7
I	15	14	13	12	18	15	9	0.91	0.87	0.82	9
J	15	14	13	12	15	13	10	0.95	0.87	0.80	10
K	14	13	13	13	15	15	13	0.93	0.90	0.90	8
L	14	13	11	10	17	16	6	0.90	0.76	0.69	15
M	14	11	11	11	17	17	11	0.83	0.80	0.80	12
N	14	11	11	10	23	15	9	0.81	0.81	0.75	14
O	13	10	10	9	19	10	7	0.79	0.75	0.71	17
P	13	13	11	11	13	13	9	1.00	0.88	0.88	11
Q	13	7	7	7	23	23	1	0.54	0.54	0.54	22
R	12	11	10	10	13	11	9	0.89	0.81	0.81	15
S	12	10	9	9	16	13	8	0.83	0.79	0.79	17
T	10	9	9	9	18	18	8	0.89	0.84	0.84	19
U	10	9	8	8	13	13	9	0.89	0.76	0.76	20
V	10	7	7	7	14	14	7	0.76	0.68	0.68	22
W	9	8	7	7	13	13	6	0.89	0.83	0.83	21
X	8	8	7	7	9	10	4	1.00	0.88	0.88	22
Y	7	6	6	6	7	7	5	0.93	0.86	0.86	25
Z	5	4	4	3	6	3	4	0.84	0.84	0.72	26

7. DETECTING ANOMALOUS CITATION GROUPS IN JOURNAL NETWORKS

[ABSTRACT]

The authors address citation cartels, groups of journals that cite each other excessively to inflate citation-based metrics. They propose the CIDRE algorithm to detect such anomalous citation groups in journal citation networks.

[DETAILED DESCRIPTION]

CIDRE (Citation Donors and REcipients) compares observed citations between journals with a null model that accounts for the journals' research-area proximity and citation volumes. Citations in excess of the null model are retained, and groups of journals connected by these excess citations are extracted as candidate citation groups with donor and recipient roles. Applied to a citation network of 48,821 journals built from the Microsoft Academic Graph, the detected groups overlap with journal groups suspended from the Journal Citation Reports. The authors suggest the algorithm can be applied to author-journal networks.

8. CITATION CLIQUES IN LOW IMPACT JOURNALS

[ABSTRACT]

This exploratory study examines how low-impact journals, defined through subject normalized Eigenfactor percentiles, are associated with denser and more reciprocating patterns of author-to-author citations. Using Crossref records, we assign journals to broad subject areas, compute subject-specific Eigenfactor scores, propagate venue quality to works and authors, match authors in low– (Case) versus high-influence (Control) venues by subject and h5, and analyze citation edges for cohesion and anomalies. Across a 10% sample of 9431 matched pairs, authors in low-impact venues exhibit significantly higher cohesion: 6.7× higher co-author citation rates and 4.7× higher reciprocity in the aggregate Case–Control comparison. A subject-aware hybrid detection pipeline flags 277 outliers with 93.5% Case purity; these outliers display an 11× clique-strength lift relative to non-outliers, revealing a stark "Two Worlds" segregation (r = 0.71) where low-impact venues operate as closed citation economies. The largest detected component (n = 23) displays a hub-and-spoke topology in which peripheral "Sycophants" funnel citations to central "Beneficiaries" through coordinated bursts, confirming a directed flow imbalance rather than reciprocal exchange among equals. Overall, cohesion, rather than broad asymmetry, accounts for the main Case–Control differences, suggesting that low-impact venues foster segregated, inward-looking citation economies that distort bibliometric indicators.

[DETAILED DESCRIPTION]

The paper uses Crossref records to assign journals to subject areas and labels venues by subject-normalised Eigenfactor percentiles, propagating venue quality to works and authors. Authors in low-impact (case) and high-impact (control) venues are matched by subject and h5-index, giving 9,431 matched pairs in a 10% sample. Author-to-author citation networks are built with features including co-author citation rate, reciprocity, clustering, k-core and centrality measures. A hybrid pipeline combining an isolation forest with a domain-specific Cohesion Composite Score flags 277 outlier authors, 93.5% of them from low-impact venues.

9. DETECTING THE MACHINE: A COMPREHENSIVE BENCHMARK OF AI-GENERATED TEXT DETECTORS ACROSS ARCHITECTURES, DOMAINS, AND ADVERSARIAL CONDITIONS

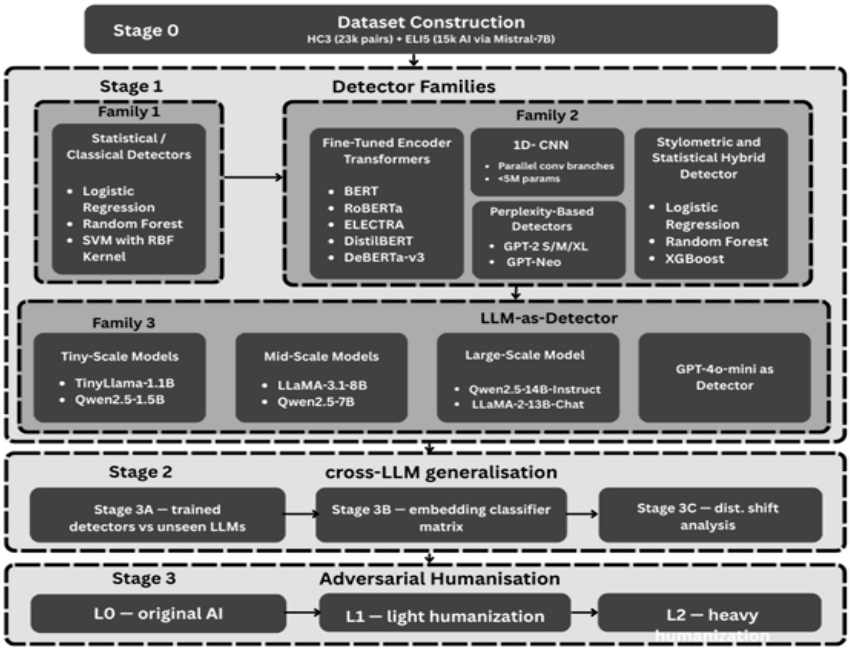
[ABSTRACT]

The rapid proliferation of large language models (LLMs) has created an urgent need for robust, generalizable detectors of machine-generated text. Existing benchmarks typically evaluate a single detector type on a single dataset under ideal conditions, leaving critical questions about cross-domain transfer, cross-LLM generalization, and adversarial robustness unanswered. This work presents a comprehensive benchmark that systematically evaluates a broad spectrum of detection approaches across two carefully constructed corpora: HC3 (23,363 paired human ChatGPT samples across five domains, 46,726 texts after binary expansion) and ELI5 (15,000 paired human–Mistral-7B samples, 30,000 texts). The approaches evaluated span classical statis tical classifiers, five fine-tuned encoder transformers (BERT, RoBERTa, ELECTRA, DistilBERT, DeBERTa-v3), a shallow 1D-CNN, a stylometric-hybrid XGBoost pipeline, perplexity-based unsupervised detectors (GPT-2/GPT-Neo family), and LLM-as-detector

prompting across four model scales including GPT-4o-mini. All detectors are further evaluated zero-shot against outputs from five unseen open-source LLMs with distributional shift analysis, and subjected to iterative adversarial humanization at three rewriting intensities (L0–L2). A principled length-matching preprocessing step is applied throughout to neutralize the well-known length confound. Our central findings are: (i) fine-tuned transformer encoders achieve near-perfect in-distribution AUROC (≥0.994) but degrade universally under domain shift; (ii) an XGBoost stylometric hybrid matches transformer in-distribution performance while remaining fully interpretable, with sentence-level perplexity coefficient of variation and AI-phrase density as the most discriminative features; (iii) LLM-as-detector prompting lags far behind fine-tuned approaches - the best open-source result is Llama-2-13b-chat-hf CoT at AUROC 0.898, while GPT-4o-mini zero-shot reaches 0.909 on ELI5 - and is strongly confounded by the generator–detector identity problem; (iv) perplexity-based detectors reveal a critical polarity inversion - modern LLM outputs are systematically lower perplexity than human text - that, once corrected, yields effective AUROC of ≈0.91; and (v) no detector generalizes robustly across LLM sources and domains simultaneously.

[DETAILED DESCRIPTION]

The classical detector family (Sections 4.1 and 13.1) computes 22 hand-crafted features, including average sentence length, sentence-length variance and standard deviation, sentence-length entropy, hedging-word density and burstiness. The stylometric hybrid (Section 13.4) adds a density of 16 characteristic AI hedge phrases and the mean, variance, standard deviation and coefficient of variation of per-sentence perplexity, low variance being treated as a sign of machine-generated text. The features feed logistic regression, random forest, SVM and XGBoost classifiers. The authors report that the interpretable stylometric hybrid matches transformer performance in-distribution.



10. HOW THE JOURNAL CHECKER TOOL WORKS (cOAlition S, Plan S Journal Checker Tool)

[ABSTRACT]

The Journal Checker Tool lets an author enter a funder, an institution and a journal and checks whether that combination offers any route to compliance with the funder's Plan S open access policy.

[DETAILED DESCRIPTION]

The tool comprises a back-end that harvests and caches data from multiple bibliographic and policy data sources, exposes an open API and executes a defined compliance algorithm. A request supplies a journal identified by its ISSN, with optional funder Crossref IDs and institution ROR IDs. The algorithm runs a separate rule-based check for each route to Plan S compliance, namely Full Open Access, Self-Archiving, Transformative Agreements and Transformative Journals, each documented as a flowchart whose transitions are logged with codes in the API logs. For each route the output states whether the combination is Compliant, Non-Compliant or Unknown, Unknown being returned when the data sources are insufficient to decide. Each result also carries qualifications guiding the user towards compliance and an audit trail recording the checks that led to the decision.

The invention also includes a supplementary credibility logic engine comprising Adaptive Decay Weighting (ADW), Cross-Signal Anomaly Detection (CSAD), Grammatical Tense Analysis (GTA), Temporal Fingerprint Evolution Tracking (TFET) and a six-phase deterministic processing pipeline, and these mechanisms were considered in the prior art search. ADW is partly disclosed by AU2012231158B2, which computes a credibility score from any number of available measures, and in principle by the Journal Checker Tool, which returns an "Unknown" result when data is insufficient. Neither, however, redistributes the weights of missing dimensions proportionally. TFET is partly disclosed by US7493403B2 and its continuation US7761583B2, which monitor a domain's ownership records over time and re-validate the domain when the records change, but neither computes a multi-dimensional drift vector against a stability threshold. The six-phase pipeline is partly disclosed by the staged processing of US7493403B2, AU2012231158B2 and the Journal Checker Tool, though none discloses the same sequence of phases or serializable state after each phase. No cited reference discloses CSAD, which amplifies risk when independent signals diverge, or GTA, which classifies the tense of a venue's indexing statements.

Search Result Links

SL. NO	SEARCH RESULT	LINK
1	US9178901B2	patents.google.com/patent/US9178901B2
2	US12113828B2	patents.google.com/patent/US12113828B2
3	US7493403B2 (continuation US7761583B2)	patents.google.com/patent/US7493403B2 ; US7761583B2
4	AU2012231158B2	patents.google.com/patent/AU2012231158B2
5	US12361214B2	patents.google.com/patent/US12361214B2
6	Schreiber (2007), Hirsch index for 26 physicists	arxiv.org/abs/0708.0120
7	Kojaku et al. (2021), Scientific Reports	doi.org/10.1038/s41598-021-93572-3
8	Spanakis et al. (2026), Citation Cliques	arxiv.org/abs/2605.11930
9	Baidya et al. (2026), Detecting the Machine	arxiv.org/abs/2603.17522
10	cOAlition S, How the Journal Checker Tool works	journalcheckertool.org/how-it-works

Disclaimer

No legal or statutory compliance advice has been sought. The report and opinion are based on technical feasibility and practical applications, and only a technical comparison of features with the searched prior art is discussed. The search was performed on third-party databases, including Google Patents, USPTO, EPO, arXiv, IEEE, ACM and publisher websites, and the authenticity, accuracy and validity of the results depend on the accuracy of the information listed in these databases. The search does not include patent applications filed in the USPTO, EPO, Indian Patent Office or other patent offices that have not yet been published, as these are not in the public domain.